

```
top - 03:59:04 up 1 day, 6:11, 1 user, load average: 0.40, 0.66, 0.76
Tasks: 138 total, 2 running, 86 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.7 us, 0.7 sy, 0.0 ni, 96.1 id, 0.0 wa, 0.0 hi, 0.7 si, 0.7 st
KiB Mem : 4048372 total, 377200 free, 1383192 used, 2287980 buff/cache
KiB Swap: 0 total, 0 free, 0 used, 2545248 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
7805	wp_d3wd+	20	0	1005380	181552	90948	R	5.0	4.5	16:14.00	php-fpm
2402	mysql	20	0	930072	389472	23396	S	2.3	9.6	287:49.28	mysqld
8	root	20	0	0	0	0	I	1.3	0.0	8:58.41	rcu_sched
1099	root	20	0	590980	20048	7952	S	0.7	0.5	6:41.25	fail2ban-server
24	root	20	0	0	0	0	S	0.3	0.0	1:18.76	ksoftirqd/3
1340	zabbix	20	0	116072	7276	5888	S	0.3	0.2	7:04.89	zabbix_agentd
3284	varnish	20	0	3582920	687308	581148	S	0.3	17.0	11:22.11	cache-main
5970	wp_d3wd+	20	0	44852	3972	3328	R	0.3	0.1	0:00.02	top
1	root	20	0	225780	9552	6672	S	0.0	0.2	0:49.46	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.27	kthreadd
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworner/0:0H
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	mm_percpu_wq
7	root	20	0	0	0	0	S	0.0	0.0	0:02.85	ksoftirqd/0
9	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_bh
10	root	rt	0	0	0	0	S	0.0	0.0	0:00.24	migration/0
11	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
12	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
13	root	rt	0	0	0	0	S	0.0	0.0	0:00.54	migration/1
14	root	20	0	0	0	0	S	0.0	0.0	0:02.99	ksoftirqd/1
16	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworner/1:0H
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/2
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.52	migration/2
19	root	20	0	0	0	0	S	0.0	0.0	0:02.88	ksoftirqd/2

Figure 2: A sample output of the 'top' command

'top' command includes information about each process, such as its process ID, the user who initiated the process, CPU time and memory usage, and the command that started the process. The output is updated in real-time, allowing for dynamic monitoring of the processes running on your system.

Interactively managing processes on your system becomes possible using the 'top' command. For instance, you can use the 'k' command to signal a process for termination, the 'r' command to change a process's priority, or the 'n' command to adjust the number of displayed processes.

The 'top' command is a powerful tool for system administrators who need to monitor the processes running on a Linux system in real-time. By using this command, you can quickly identify processes that consume excessive CPU or memory resources, take appropriate actions to manage them effectively, and ensure the smooth and efficient operation of your system.

Important options of the 'top' command

Several options can be used with the 'top' command to customise the output and display additional information. One particularly useful option is '-d', which allows you to specify the interval between updates. By default, the 'top' command updates the output every 3 seconds. However, you can use the '-d' option to specify a different update interval. For example, to update the output every 5 seconds, you would type the following at the prompt:

```
$top -d 5
```

Another useful option is '-p', which enables you to specify the process IDs of the processes you want to monitor. This is useful if you want to monitor specific processes rather

than all the processes running on your system. For example, if you want to monitor processes with PIDs 1234 and 5678, you would type the following at the command prompt:

```
$top -p 1234,5678
```

You can also use the 'z' command to toggle between displaying all the processes running on your system and only those currently using CPU time. This is useful when you want to focus on the processes that consume the most CPU time and ignore the idle ones.

How to use the 'top' command in batch mode

By default, the 'top' command displays a dynamic view of the processes running on your system, sorted by the amount of CPU they are using. However, you can also use the 'top' command in batch mode to generate a static report of the processes running on your system. To do so, you need to use the '-b' option. This option tells the 'top' command to generate a static report of the processes running on your system, rather than displaying a dynamic view. For example, if you want to generate a report of the processes running on your system and save it to a file called 'processes.txt', use the following command:

```
$top -b -n 1 > processes.txt
```

The '-n' option is used to specify the number of iterations the 'top' command should run before generating the report. In this example, we are using the '-n 1' option to tell the 'top' command to run for one iteration before generating the report. When you run the 'top' command in batch mode, the output is saved to a file in plain text format. You can then open the file in a text editor or import it into a spreadsheet program for further analysis. Finally, type 'q' to exit the 'top' command at any point in time.

In conclusion, the 'top' command can be used in batch mode to generate a static report of the processes running on a Linux system in plain text format, which can be easily analysed and shared with others.

Other commands equivalent to 'top'

There are other commands similar to 'top' that can be used to achieve similar results. Here are some of the most commonly used alternatives.

'htop': 'htop' is a more advanced version of the 'top' command that provides a more user-friendly interface and allows you to sort processes by various criteria, search for specific processes, and view detailed information about each process.